

AWS checklist and maintenance guide

Station name	TAS_A
People present	RSE, SIBA instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency
Date	13/8-2015
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %: 20

Notes:

arrival 13:40 Very wet.
leave ~ 15:

Snow height?

15 cm

15 "

15 "

25 "

18 "

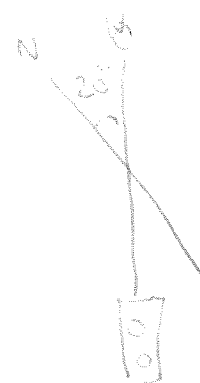
Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	+ 20	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)		Y / ☒ N
Download data to PC and/or change CF Card (wait for green light)		☒ Y / N
Name of logger program	2014d	
Photo or screen dump of values in fast scan mode		☒ Y / N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		☒ Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	0	°
Mast tilt across boom looking from radiometer (- if clockwise)	-2	°
Boom direction relative to north		° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.		☒ Y / N
Radiometer aligned with mast and inclinometer? If not, measure.		☒ Y / N
Temperature / humidity height (bottom of casing – surface)	235	cm
Sonic ranger on AWS height (membrane – surface)	245	cm
Wind sensor height (center of propeller – surface)	287	cm
Sonic ranger on stakes height (membrane – surface)	186	cm
Free length of stake – outer, holding boom with sonic ranger	205	cm
Free length of stake – middle	215	cm
Free length of stake – outer, holding boom without sonic ranger	210	cm
Free length of ablation hose on/above the ice surface	1111	cm
Alt: depth of ablation hose using markings on hose	0	cm
Height of ablation hose fluid level above the ice surface	116	cm
Vertical difference in surface height (at station – at hose) (approx)	0	cm
Length of thermistor string on surface from surface marking	-	cm
Height of surface irregularities	10	cm
Stake assembly still standing?	☒ Y / N / more or less	
Apparent damage to sensors? If yes, which?		Y / ☒ N

210
snow covered
covered
in snow
water

Notes:

SPA sensor på skrå revet i stykke
fin strøm på stationen



Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	080024		- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	01355	SN00052	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	GEUS34	—	
Wind sensor	119282	—	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly	1012012	—	
T/hum assembly casing			- Fan spins and sounds OK: Y/N
T/hum assembly bracket			
Temperature probe in T/hum assembly			
HygroClip in T/hum assembly	61180280	—	- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS	4110	6263	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel	030		- Output OK: Y/N - Clean: Y/N
GPS antenna	GEUS05	GEUS22 33-4027-00-3000	- Stuck to top of logger enclosure: Y/N
Data logger	AWS 1/209	—	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader	8325	—	- LED green or flashing orange (active): Y/N
Multiplexer	22106 E320	—	- Clicking as it should: Y/N
Iridium modem IMEI number	3002 3401 1039 970	—	- Transmissions tested: Y/N - Problems: Y/N
Barometer	3696480	—	
Sonic ranger on stakes	4284	2080	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string	Under snow	M03	- Old string left in ice: Y/N/cut
Ablation hose assembly	Under snow		- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?		Y / N
Name of logger program	Promax 2015 e	
Photo or screen dump of values in fast scan mode		Y / N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		Y / N
Mast tilt in boom direction (+ if radiometer tilting down)		°
Mast tilt across boom looking from radiometer (- if clockwise)		°
Boom direction relative to north (radiometer should point to true south)	90° true/magn	
Box on wind sensor exactly towards mast? If not, change.		Y / N
Radiometer aligned with mast and inclinometer? If not, align.		Y / N
Temperature / humidity height (bottom of casing – surface)		cm
Sonic ranger on AWS height (membrane – surface)		cm
Wind sensor height (center of propeller – surface)		cm
Sonic ranger on stakes height (membrane – surface)		cm
Free length of stake – outer, holding boom with sonic ranger	205	cm
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Free length of ablation hose on/above the ice surface		cm
Alt: depth of ablation hose using markings on hose		cm
Height of ablation hose fluid level above the ice surface	116	cm
Vertical difference in surface height (at station – at hose) (approx)	0	cm
Length of thermistor string on surface from surface marking	under snow	cm
Height of surface irregularities	under snow	cm

Notes: